

Operating System: System Calls



System Calls

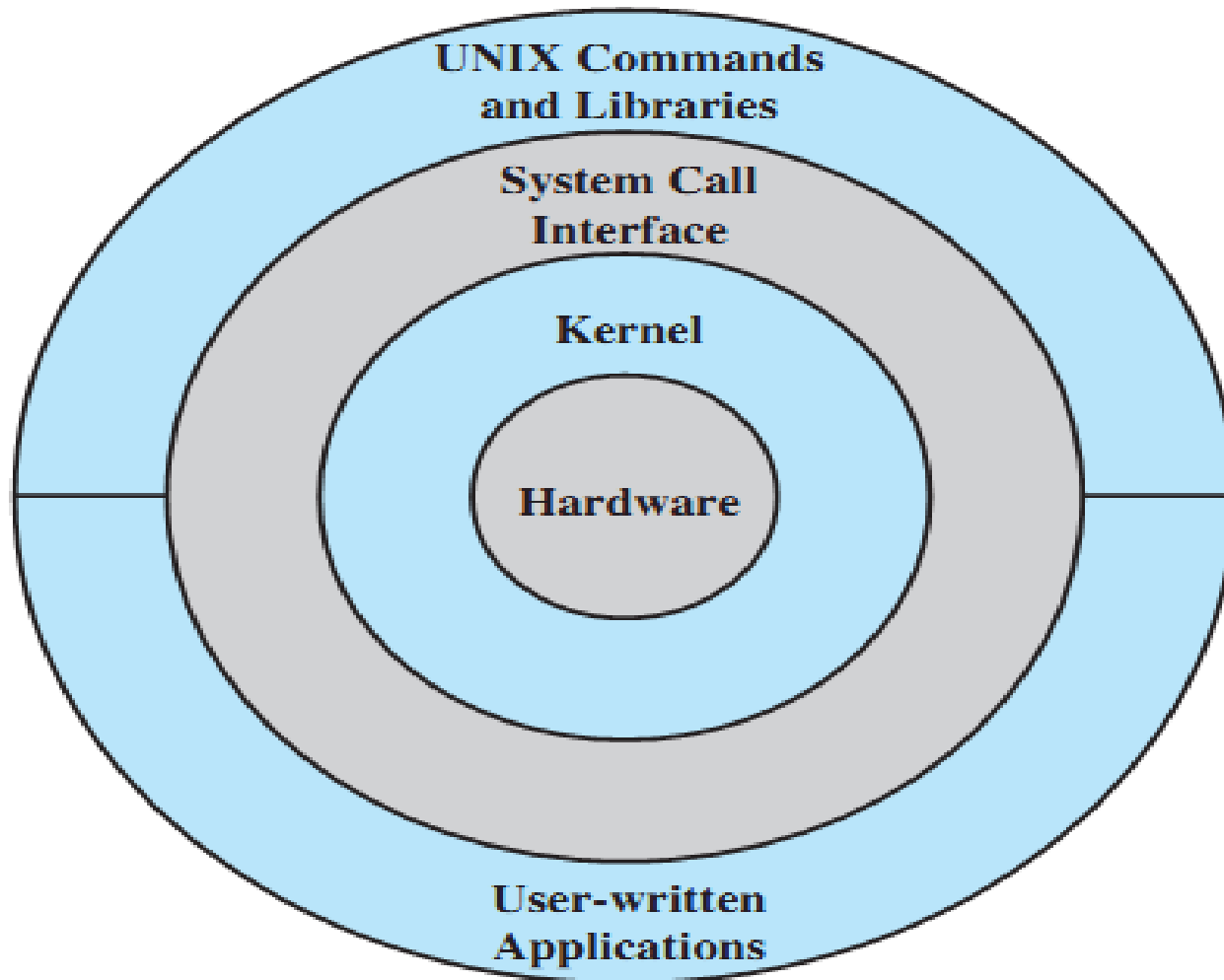
- Programming interface to the services provided by the OS
- Typically written in a high-level language (C or C++)
- Mostly accessed by programs via a high-level **Application Programming Interface (API)** rather than direct system call use
- Three most common APIs are Win32 API for Windows, POSIX API for POSIX-based systems (including virtually all versions of UNIX, Linux, and Mac OS X), and Java API for the Java virtual machine (JVM)

(Note that the system-call names used throughout this text are generic)





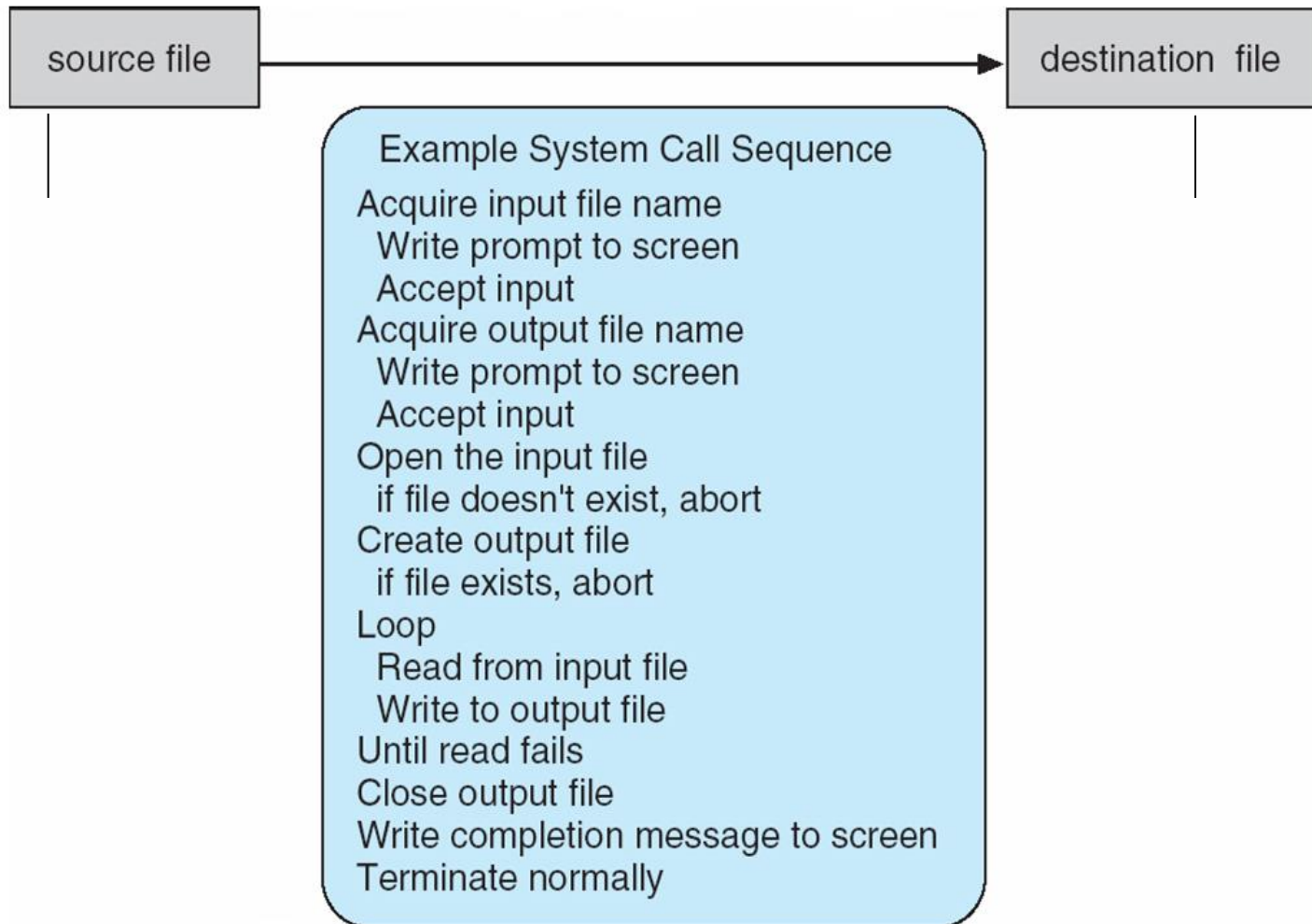
General UNIX Architecture (1)



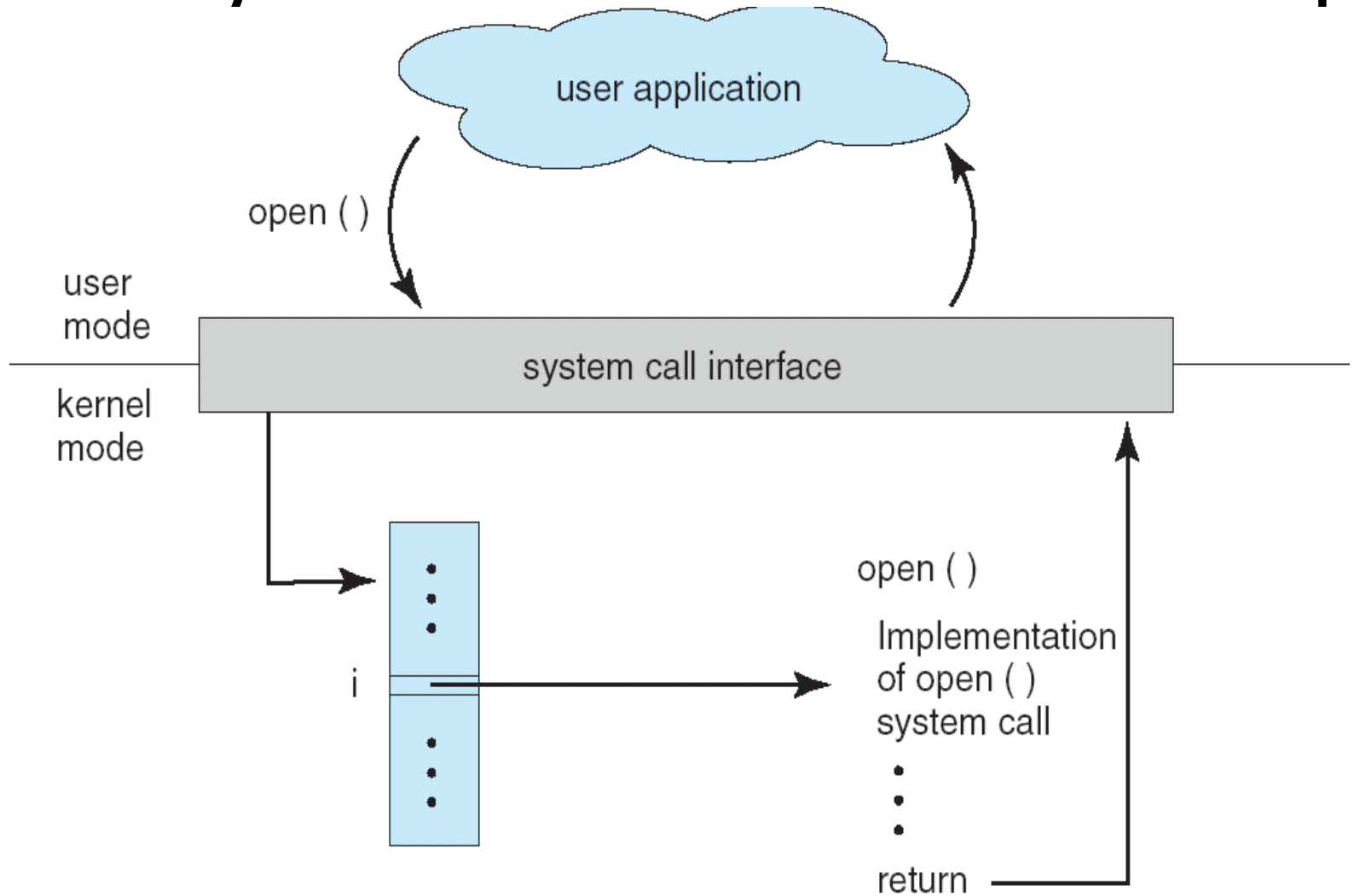


Example of System Calls

- System call sequence to copy the contents of one file to another file

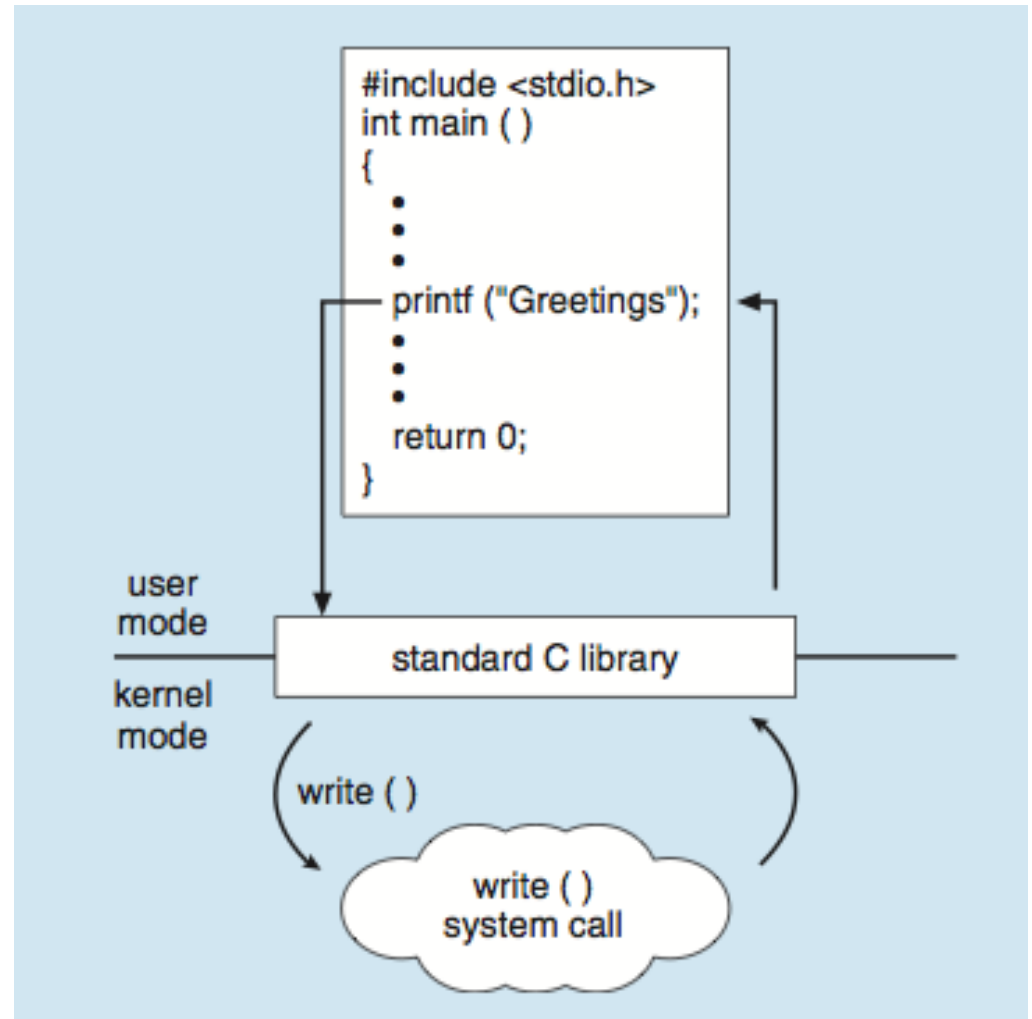


API – System Call – OS Relationship



Standard C Library Example

- C program invoking printf() library call, which calls write() system call



Types of System Calls

PIC FDP

- Process control
- File management
- Device management
- Information maintenance
- Communications
- Protection



Types of System Calls

■ Process control

- end, abort
- load, execute
- create process, terminate process
- get process attributes, set process attributes
- wait for time
- wait event, signal event
- allocate and free memory

- Dump memory if error
- **Debugger** for determining **bugs, single step** execution
- **Locks** for managing access to shared data between processes





Types of System Calls

■ File management

- create file, delete file
- open, close file
- read, write, reposition
- get and set file attributes

■ Device management

- request device, release device
- read, write, reposition
- get device attributes, set device attributes
- logically attach or detach devices





Types of System Calls (Cont.)

■ Information maintenance

- get time or date, set time or date
- get system data, set system data
- get and set process, file, or device attributes

■ Communications

- create, delete communication connection
- send, receive messages if **message passing model** to **host name** or **process name**
 - ▶ From **client** to **server**
- **Shared-memory model** create and gain access to memory regions
- transfer status information
- attach and detach remote devices





Types of System Calls (Cont.)

■ Protection

- Control access to resources
- Get and set permissions
- Allow and deny user access





Examples of Windows and Unix System Calls

	Windows	Unix
Process Control	CreateProcess() ExitProcess() WaitForSingleObject()	fork() exit() wait()
File Manipulation	CreateFile() ReadFile() WriteFile() CloseHandle()	open() read() write() close()
Device Manipulation	SetConsoleMode() ReadConsole() WriteConsole()	ioctl() read() write()
Information Maintenance	GetCurrentProcessID() SetTimer() Sleep()	getpid() alarm() sleep()
Communication	CreatePipe() CreateFileMapping() MapViewOfFile()	pipe() shmget() mmap()
Protection	SetFileSecurity() InitializeSecurityDescriptor() SetSecurityDescriptorGroup()	chmod() umask() chown()

